Nanyang Technological University, Singapore

EE6407 Genetic Algorithms and Machine Learning

Assignments 1 and 2

Naïve Bayes and Fisher linear discriminant classifiers

Submitted by

Han Chang G2602636F

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1 Assignment 1

The training set (`TrainingData.xlsx`) has 120 samples with four features A–D in columns A–D and a class label $\omega \in \{1, 2, 3\}$ in column E. The test set (`TestData.xlsx`) has 30 unlabelled samples. Rows are numbered by spreadsheet row; neither file has a header row.

1.1 Missing values and outliers

The training set does contain missing values, three of them, each a "?" typed where a number should be:

Table 1: Incomplete training rows.

Row	Missing feature	Class	Remaining values (A, B, C, D)
39	A	2	(–, 3.3, 5.7, 2.5)
61	A	1	(–, 3.6, 1.0, 0.2)
83	D	3	(5.9, 3.0, 4.2, –)

Outliers are there as well, three of them. Each class has to be inspected on its own, because the three classes occupy very different ranges: feature C is about 1.5 in class 1 and about 5.5 in class 2. I screened with Tukey's far-out fence, class by class and feature by feature, so a value is flagged when it falls outside

$$\big[\, Q_1 - 3 \cdot \mathrm{IQR},\ Q_3 + 3 \cdot \mathrm{IQR} \,\big], \qquad \mathrm{IQR} = Q_3 - Q_1,$$

where Q_1 and Q_3 are that class's lower and upper quartiles of that feature. Quartiles are interpolated linearly between order statistics, the convention of `numpy.percentile` and `statistics.quantiles(method='inclusive')`. The data are recorded to one or two decimal places, so the fences are computed in exact decimal arithmetic: a value sitting exactly on a fence should not be flagged because of floating-point noise. The 3·IQR rule is a conservative screening criterion and not a definition of what an outlier is, and the 1.5·IQR variant is discussed below.

The screening runs in a fixed order. The three incomplete rows are set aside first, not imputed and not used in the fence calculation. For each class and each of the four features, Q_1, Q_3 and the far-out fence are then computed on the remaining rows, every cell outside its fence is flagged, and the union of all rows holding a flagged cell is removed.

Three cells are flagged, in three different rows, so three cells and three samples are removed. The two counts coincide only because no row contains more than one flagged cell:

Table 2: Far-out outliers. μ and σ are the class's mean and standard deviation of that feature after the three rows are removed; $z = (\text{value} - \mu)/\sigma$.

Row	Sample (A, B, C, D)	Class	Flagged value	μ (σ)	z	3·IQR fence
91	(6.3, 3.3, 16, 2.5)	2	$C = 16$	5.551 (0.586)	17.8	[2.70, 8.30]
96	(10.6, 3.2, 1.4, 0.2)	1	$A = 10.6$	5.005 (0.328)	17.1	[3.80, 6.25]
117	(5.7, 2.9, 4.2, 5.3)	3	$D = 5.3$	1.344 (0.196)	20.2	[0.30, 2.40]

The other three features of each of these rows sit inside their own fences, so each row is a single-cell anomaly and not a wholly unusual sample. The flagged values are 17 to 20 standard deviations from their class mean, well beyond anything else the class produces. A z-score rule with the same class statistics ($|z| > 3$) flags exactly the same three cells, so the conclusion does not depend on the rule chosen. Why the entry is wrong cannot be known from the data: a reading 17σ out is consistent with a recording error, but the intended value is unrecoverable, so these rows are dropped rather than corrected.

The milder 1.5·IQR rule flags eight more borderline cells (rows 1, 26, 69, 79, 97, 98, 105, 116), with $|z|$ between 2.12 and 2.98. They sit just outside the usual range for their class, and nothing in the data suggests an error, so I kept them. That is a judgement and not a finding: a different threshold would keep or drop a different set, which is why the threshold and the quartile convention are stated at all.

Three methods can address the missing entries. Listwise deletion drops the incomplete samples; it invents nothing, but it is unbiased only when the values are missing completely at random, which three cells cannot establish. With 3 of 120 samples affected it is defensible here. Class-conditional imputation fills the entry with the mean or median of that feature over samples of the same class; it keeps the sample but shrinks the feature's variance slightly, and it needs a known label, so an unconditional mean would have to be used on the unlabelled test data. Model-based imputation predicts the entry from the other features, by k-NN averaging or by regression on the observed ones, with EM as the likelihood-based alternative; it exploits the correlation between features (A and C, 0.875) but need not recover the true entry.

Three methods can address the outliers. Detection and removal flags values by a rule, a z-score beyond ± 3 or the IQR fences above, and deletes those samples; it works when the outliers are few and evidently wrong, and it is what I applied here. Capping, or winsorising, replaces a value outside the fence with the nearest fence value or with a chosen percentile such as the 1st or 99th, keeping the sample but limiting its influence. The outlier can also be treated as missing and imputed (class median or k-NN), or kept while the parameters are estimated with robust statistics (median and MAD), so one extreme value cannot dominate the fit.

1.2 Naïve Bayes classifier design

As instructed, the 3 samples with missing values and the 3 outlier samples are removed, which leaves 114 training samples (class 1: 38, class 2: 37, class 3: 39).

Each sample is a point $\mathbf{x} = (x_1, x_2, x_3, x_4) = (A, B, C, D)$, and the Bayes decision rule assigns it to the class with the largest posterior probability

$$P(\omega_i \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \omega_i)\, P(\omega_i)}{p(\mathbf{x})}, \qquad i = 1, 2, 3.$$

Because $p(\mathbf{x})$ is the same for every class, it is enough to compare $p(\mathbf{x} \mid \omega_i)P(\omega_i)$.

Naïve Bayes then assumes that, given the class, the four features are independent, so

$$p(\mathbf{x} \mid \omega_i) = \prod_{j=1}^{4} p(x_j \mid \omega_i).$$

This replaces one 4-dimensional density with four 1-dimensional ones, far easier to estimate from 114 samples. It is an approximation: A and C correlate at 0.875 overall, and at 0.33, 0.89 and 0.74 within the three classes, so the joint density is not a product of marginals. The classifier can still be accurate, because a decision only needs the ranking of the three scores.

The features are continuous, so each one needs a density within each class. I took the Gaussian form because a mean and a variance per class-feature pair specify it completely. Continuity alone does not make a variable Gaussian, and no normality check is made here:

$$p(x_j \mid \omega_i) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} \exp\left(-\frac{(x_j - \mu_{ij})^2}{2\sigma_{ij}^2} \right).$$

With N_i the number of class-i samples and $N = 114$, the parameters are estimated as

$$P(\omega_i) = \frac{N_i}{N}, \qquad \mu_{ij} = \frac{1}{N_i} \sum_{k \in \omega_i} x_{kj}, \qquad \sigma_{ij}^2 = \frac{1}{N_i - 1} \sum_{k \in \omega_i} (x_{kj} - \mu_{ij})^2.$$

Table 3: Fitted parameters of the three class-conditional Gaussians (114 samples after cleaning).

Class	Prior $P(\omega_i)$	A		B		C		D	
		μ	σ^2	μ	σ^2	μ	σ^2	μ	σ^2
1	$38/114 = 0.3333$	5.0053	0.1075	3.4079	0.1472	1.4789	0.0293	0.2553	0.0133
2	$37/114 = 0.3246$	6.5541	0.4414	2.9459	0.0842	5.5514	0.3437	2.0108	0.0715
3	$39/114 = 0.3421$	5.9615	0.2535	2.7821	0.0968	4.3077	0.2328	1.3436	0.0383

The unbiased variance $(N_i - 1)$ is used.

Taking the natural logarithm of $p(\mathbf{x} \mid \omega_i)P(\omega_i)$ turns the product into a sum and does not change which class is largest:

$$g_i(\mathbf{x}) = \ln P(\omega_i) - \sum_{j=1}^{4} \left[\tfrac{1}{2} \ln(2\pi\sigma_{ij}^2) + \frac{(x_j - \mu_{ij})^2}{2\sigma_{ij}^2} \right], \qquad i = 1, 2, 3.$$

The term $\frac{1}{2} \cdot 4 \cdot \ln 2\pi = 3.6758$ is the same for all three classes, so it shifts the scores but never the ranking. The worked example below gives the score both with and without it.

The rule is to assign $\mathbf{x}$ to ω_k where $k = \arg\max_i g_i(\mathbf{x})$, with a tie broken towards the smaller class index (no test sample ties). Each g_i is quadratic in $\mathbf{x}$ with a diagonal covariance matrix, so the decision boundaries are quadratic surfaces in four dimensions. On the 114 cleaned training samples the classifier is 95.6% correct, which is a resubstitution figure and therefore optimistic, since every sample helped fit the parameters that then classify it:

Table 4: Naïve Bayes confusion matrix on the 114 cleaned training samples (rows = true class, columns = predicted class).

True	$\to 1$	$\to 2$	$\to 3$	Recall
1	38	0	0	$38/38 = 100\%$
2	0	34	3	$34/37 = 91.9\%$
3	0	2	37	$37/39 = 94.9\%$

All five errors are confusions between classes 2 and 3: rows 12, 23 and 85 (class 2 read as class 3) and rows 56 and 89 (class 3 read as class 2). Class 1 is never confused with anything, which is what its well-separated C and D values predict (Figure 1).

Because 95.6% is measured on the samples used for fitting, a held-out estimate is also reported. Refitting the classifier 114 times, each time on 113 samples and testing on the omitted one (leave-one-out), gives 94.7%:

Table 5: Leave-one-out confusion matrix for the same classifier (each row refits without that sample).

True	$\to 1$	$\to 2$	$\to 3$
1	38	0	0
2	0	34	3
3	0	3	36

Only one prediction changes (row 56 becomes an error, giving $108/114 = 94.7\%$), because its neighbours then supply slightly different variances. The model has only $2{+}12{+}12 = 26$ free parameters for 114 samples, which is why the two figures are so close.

As a worked example, take test sample 11, $\mathbf{x} = (6.0, 2.2, 5.0, 1.5)$. Its per-feature penalty terms $\frac{1}{2}\ln(2\pi\sigma^2) + (x - \mu)^2/(2\sigma^2)$ come out as follows:

Table 6: Class scores for test sample 11. $\tilde{g}$ is the reduced discriminant (without the 3.6758 constant); P is the posterior of the class.

Class	$\ln P(\omega_i)$	A	B	C	D	$g_i(\mathbf{x})$	$\tilde{g}_i(\mathbf{x})$	$P(\omega_i \mid \mathbf{x})$
1	−1.0986	4.4046	4.9158	210.9042	56.7900	−278.11	−274.44	0.00%
2	−1.1253	0.8578	2.9853	0.8272	1.4237	−7.22	−3.54	2.71%
3	−1.0726	0.2356	1.5016	1.2195	−0.3928	**−3.64**	**0.04**	**97.29%**

For example, for class 3 and feature A: $\frac{1}{2}\ln(2\pi \times 0.2535) + (6.0 - 5.9615)^2/(2 \times 0.2535) = 0.2327 + 0.0029 = 0.2356$. The four terms sum to 2.564, so $g_3 = -1.0726 - 2.564 = -3.64$, the largest of the three scores: sample 11 lands in class 3. Class 1 loses by a wide margin, mostly because $C = 5.0$ is far from class 1's mean of 1.48 and its variance is small (0.0293), so the squared distance is divided by a small number.

1.3 Predicted class labels of the test data

Table 7: Discriminant values and predictions for the 30 test samples. P is the posterior probability of the predicted class; margin is the gap between the best and the second-best score.

Test #	A	B	C	D	g_1	g_2	g_3	Predicted	P (%)	margin
1	4.8	3.1	1.6	0.2	0.34	−50.56	−36.53	**1**	100.0	36.87
2	5.1	3.5	1.4	0.3	0.97	−51.05	−37.03	**1**	100.0	38.00
3	4.8	3.0	1.4	0.1	−0.55	−55.39	−41.78	**1**	100.0	41.23
4	5.7	3.8	1.7	0.3	−2.45	−48.49	−34.84	**1**	100.0	32.39
5	5.5	4.2	1.4	0.2	−2.27	−59.89	−46.57	**1**	100.0	44.30
6	5.0	3.4	1.5	0.2	1.10	−52.06	−38.33	**1**	100.0	39.43
7	4.6	3.1	1.5	0.2	0.01	−52.56	−38.71	**1**	100.0	38.72
8	5.8	4.0	1.2	0.2	−4.35	−59.00	−46.06	**1**	100.0	41.71
9	5.0	3.6	1.4	0.2	0.88	−54.57	−41.04	**1**	100.0	41.91
10	4.6	3.4	1.4	0.3	0.28	−52.38	−38.53	**1**	100.0	38.81
11	6.0	2.2	5.0	1.5	−278.11	−7.22	−3.64	**3**	97.3	3.58
12	6.5	3.2	5.1	2.0	−347.27	−1.99	−8.98	**2**	99.9	6.99
13	7.7	3.0	6.1	2.3	−554.42	−3.83	−25.58	**2**	100.0	21.75
14	6.9	3.2	5.7	2.3	−476.52	−2.44	−19.27	**2**	100.0	16.84
15	7.9	3.8	6.4	2.0	−565.89	−8.73	−28.32	**2**	100.0	19.59
16	6.5	3.0	5.2	2.0	−360.23	−1.50	−8.69	**2**	99.9	7.18
17	6.2	2.8	4.8	1.8	−284.42	−2.70	−3.89	**2**	76.6	1.19
18	6.7	2.5	5.8	1.8	−423.21	−2.91	−9.52	**2**	99.9	6.61
19	6.2	3.4	5.4	2.3	−424.60	−3.29	−17.12	**2**	100.0	13.83
20	6.5	3.0	5.5	1.8	−375.26	−1.64	−7.12	**2**	99.6	5.49
21	6.8	3.0	5.5	2.1	−417.93	−1.45	−12.69	**2**	100.0	11.24
22	5.8	2.7	4.1	1.0	−141.53	−12.51	−2.25	**3**	100.0	10.26
23	6.8	2.8	4.8	1.4	−252.47	−4.93	−2.49	**3**	92.0	2.44
24	6.7	3.1	4.4	1.4	−207.27	−6.00	−2.19	**3**	97.8	3.81
25	5.2	2.7	3.9	1.4	−149.85	−10.31	−2.11	**3**	100.0	8.20
26	5.7	2.6	3.5	1.0	−93.78	−16.10	−3.78	**3**	100.0	12.32
27	5.6	2.5	3.9	1.1	−130.06	−13.28	−2.33	**3**	100.0	10.94
28	5.5	2.5	4.0	1.3	−152.15	−10.77	−1.59	**3**	100.0	9.18
29	5.0	2.0	3.5	1.0	−96.05	−22.61	−8.46	**3**	100.0	14.15
30	6.4	3.2	4.5	1.5	−221.88	−5.14	−2.22	**3**	94.9	2.93

So the 30 test samples fall into three groups: samples 1 to 10 go to class 1, samples 12 to 21 to class 2, and sample 11 with samples 22 to 30 to class 3.

Class 1 is separated from the other two by a wide margin, mostly through features C and D. The

closest decisions are sample 17 ($g_2 - g_3 = 1.19$, so $P(\omega_2 \mid \mathbf{x}) = 76.6\%$), sample 23 (margin 2.44, $P = 92.0\%$) and sample 30 (margin 2.93, $P = 94.9\%$); the worked example, sample 11, is only the fourth closest (margin 3.58). All three of the closest samples lie where classes 2 and 3 overlap in C, which is visible in Figure 1. These are model scores; with no test labels, nothing here measures accuracy.

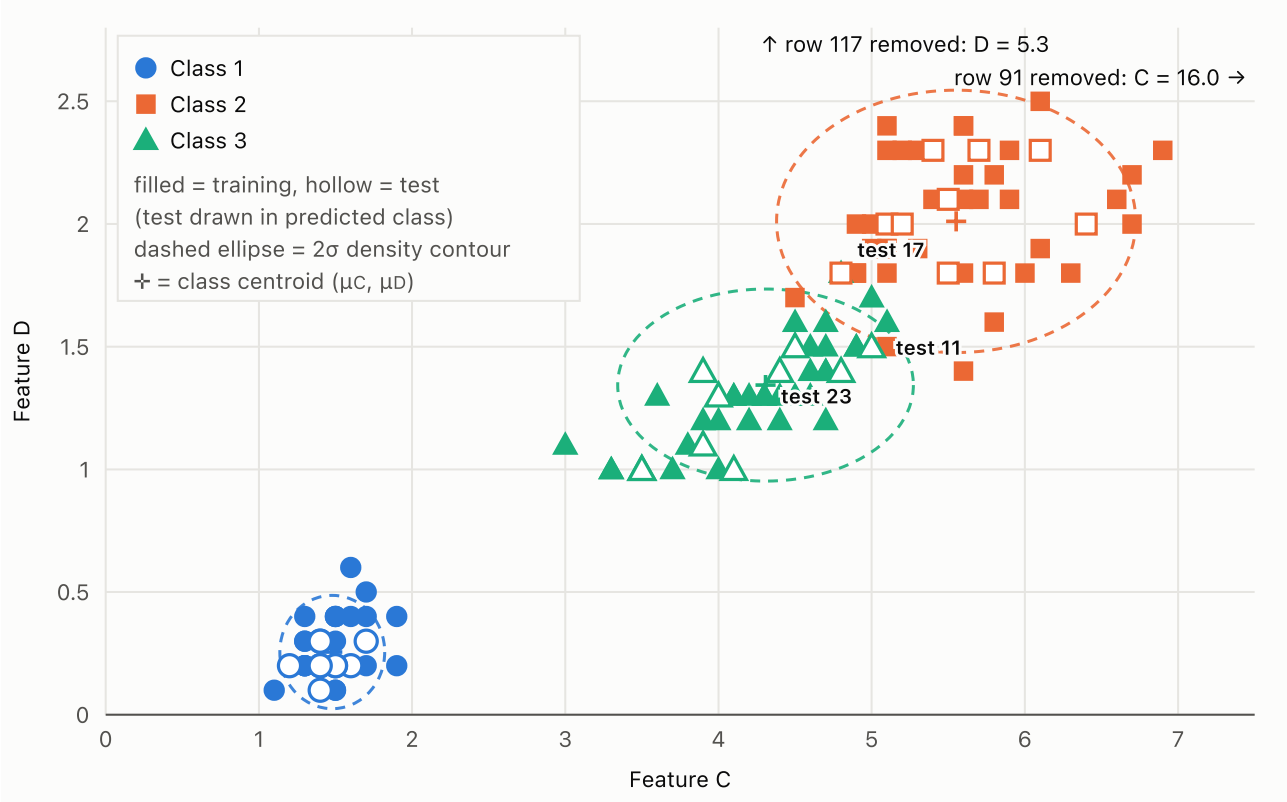


Figure 1: Features C and D for the 114 cleaned training samples (filled) and the 30 test samples (hollow, drawn in their predicted class), with the 2σ density contour of each fitted class Gaussian and its centroid marked by a cross. Class 1 is well separated; classes 2 and 3 overlap, and that overlap is where the closest test decisions (17, 23, 30) and the five training errors lie. Row 91's outlier ($C = 16.0$) and row 117's ($D = 5.3$) fall outside the plotted range, and row 96's outlier is in feature A.

2 Assignment 2

The training set (`data_train.mat`) has 577 samples with five features x_1–x_5; `label_train.mat` gives each sample's class as $+1$ or -1. There are 301 samples labelled $+1$ and 276 labelled -1, and no missing values. The test set (`data_test.mat`) has 23 unlabelled samples. Below, class ω_1 means label $+1$ and class ω_2 means label -1.

2.1 Fisher linear discriminant classifier

Fisher's method looks for a direction $\mathbf{w}$ such that, once every sample is projected onto it ($y = \mathbf{w}^T\mathbf{x}$), the two projected class means are far apart while each class's projected samples stay close together. It maximises the Fisher criterion

$$J(\mathbf{w}) = \frac{(\tilde{m}_1 - \tilde{m}_2)^2}{\tilde{s}_1^2 + \tilde{s}_2^2} = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}},$$

where $\tilde{m}_i$ is the projected class mean, $\tilde{s}_i^2 = \sum_{y\in\omega_i}(y - \tilde{m}_i)^2$ is the projected within-class *scatter* (a sum of squares, not divided by a count), $S_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$ is the between-class scatter matrix, and S_W is the within-class scatter matrix defined below.

The class mean vectors are $\mathbf{m}_i = (1/N_i)\sum_{\mathbf{x}\in\omega_i}\mathbf{x}$, with $N_1 = 301$ and $N_2 = 276$:

$$\begin{aligned}
\mathbf{m}_1 &= (-0.0418,\ 0.0671,\ -0.0540,\ 0.0912,\ -0.0147)^T,\\
\mathbf{m}_2 &= (1.0428,\ 2.0623,\ 0.9408,\ 1.2551,\ 1.6047)^T,\\
\mathbf{m}_1 - \mathbf{m}_2 &= (-1.0846,\ -1.9952,\ -0.9947,\ -1.1639,\ -1.6194)^T.
\end{aligned}$$

The within-class scatter of each class is $S_i = \sum_{\mathbf{x}\in\omega_i}(\mathbf{x}-\mathbf{m}_i)(\mathbf{x}-\mathbf{m}_i)^T$, again a scatter and not divided by N_i, and $S_W = S_1 + S_2$:

Table 8: Within-class scatter matrix S_W (5×5, symmetric).

568.11	16.99	−0.13	21.12	−9.17
16.99	567.74	−17.44	28.77	−8.01
−0.13	−17.44	540.03	15.81	−8.42
21.12	28.77	15.81	604.84	−8.36
−9.17	−8.01	−8.42	−8.36	637.31

S_W is non-singular, being strongly diagonally dominant, so the system $S_W\mathbf{w} = \mathbf{m}_1 - \mathbf{m}_2$ has a unique solution and can be solved directly. The script does that by Gaussian elimination, without forming an explicit inverse.

Setting the gradient of $J(\mathbf{w})$ to zero gives the generalised eigenvalue problem $S_B\mathbf{w} = \lambda S_W\mathbf{w}$. Because $S_B\mathbf{w}$ always points along $\mathbf{m}_1 - \mathbf{m}_2$, and the length of $\mathbf{w}$ does not change the decision, the solution is

$$\mathbf{w} = S_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2) = 10^{-3} \times (-1.7860,\ -3.4729,\ -1.9467,\ -1.6826,\ -2.6581)^T.$$

The entries are small only because S_W is a sum over 577 samples; scaling $\mathbf{w}$ and w_0 by the same positive constant gives the same decision. This orientation ($\mathbf{m}_1 - \mathbf{m}_2$) places class ω_1 on the positive side of the boundary. At this $\mathbf{w}$, $J(\mathbf{w}) = 0.0171$: the projected means are $\tilde{m}_1 = -0.168 \times 10^{-3}$ and $\tilde{m}_2 = -17.233 \times 10^{-3}$ (8.53×10^{-3} either side of the boundary), and the projected within-class standard deviations are 5.19×10^{-3} and 5.72×10^{-3}. The second class is the more spread out after projection, which is consistent with its larger share of the training errors (26 against 13).

Fisher's criterion fixes only the direction, so the bias is a separate convention. The one I used puts the boundary through the midpoint of the two class means, i.e. $g((\mathbf{m}_1 + \mathbf{m}_2)/2) = 0$:

$$w_0 = -\mathbf{w}^T\frac{\mathbf{m}_1 + \mathbf{m}_2}{2} = -\frac{(-0.1678) + (-17.2331)}{2} \times 10^{-3} = 8.7005 \times 10^{-3}.$$

The midpoint assumes equal priors and equal misclassification costs. It is a choice rather than a tuned threshold: tuning the bias on the training data could shrink the 26 errors against 13, and nothing here is tuned on the test set.

Weights and bias together then give the discriminant function

$$\begin{aligned} g(\mathbf{x}) &= \mathbf{w}^T\mathbf{x} + w_0 \\ &= 10^{-3} \times \big(-1.7860\,x_1 - 3.4729\,x_2 - 1.9467\,x_3 - 1.6826\,x_4 - 2.6581\,x_5 + 8.7005\big). \end{aligned}$$

If $g(\mathbf{x}) > 0$ the sample goes to ω_1 (label $+1$), otherwise to ω_2 (label -1), so an exact tie $g(\mathbf{x}) = 0$ falls to ω_2. The boundary $g(\mathbf{x}) = 0$ is a hyperplane in the 5-dimensional feature space. On the 577 training samples the classifier is 93.2% correct:

Table 9: Fisher discriminant confusion matrices on the 577 training samples (rows = true label, columns = predicted label).

True	Resubstitution		Leave-one-out	
	$+1$	-1	$+1$	-1
$+1$	288	13	287	14
-1	26	250	26	250

Resubstitution gives $538/577 = 93.24\%$ and leave-one-out $537/577 = 93.07\%$. With only six free parameters (five weights and a bias) the two agree closely, the single extra leave-one-out error being in class ω_1.

2.2 Predicted class labels of the test data

Test sample 1, $\mathbf{x} = (0.0058, 1.5626, 0.4474, 2.0909, 0.8973)^T$, has products $w_j x_j$ ($\times 10^{-3}$) of -0.0103, -5.4267, -0.8709, -3.5181 and -2.3852, so $\mathbf{w}^T\mathbf{x} = -12.2113 \times 10^{-3}$ and $g(\mathbf{x}) = (-12.2113 + 8.7005) \times 10^{-3} = -3.511 \times 10^{-3} < 0$, which gives label -1.

In the end test samples 1 to 9 get label -1 (class ω_2) and test samples 10 to 23 get label $+1$ (class ω_1).

Every test sample has $|g(\mathbf{x})|$ between 3.2×10^{-3} and 4.4×10^{-3}, i.e. 0.60 to 0.81 feature units from the boundary, so none of the decisions is close. They do fall where the two training classes overlap (Figure 2), which is where the 39 training errors are. The test labels are unavailable, so the training figures are the only accuracy measure.

Table 10: Discriminant values and predictions for the 23 test samples. Inputs are shown rounded to 3 decimal places; $g(\mathbf{x})$ is computed from the unrounded data.

Test #	x_1	x_2	x_3	x_4	x_5	$g(\mathbf{x}) \times 10^3$	Predicted label
1	0.006	1.563	0.447	2.091	0.897	−3.511	**−1**
2	1.229	1.440	0.878	−0.119	1.469	−3.907	**−1**
3	2.340	0.968	−0.342	2.123	0.848	−4.002	**−1**
4	0.232	1.969	0.359	1.583	0.804	−4.051	**−1**
5	0.458	2.233	−0.623	1.563	0.725	−3.217	**−1**
6	0.697	2.393	−1.182	0.833	1.532	−4.028	**−1**
7	−0.174	1.710	0.617	1.233	1.335	−3.750	**−1**
8	1.171	1.300	0.185	1.828	0.785	−3.428	**−1**
9	2.065	0.830	0.279	1.601	1.032	−3.854	**−1**
10	0.835	0.930	−0.577	0.649	0.183	3.525	**+1**
11	0.495	1.605	0.416	0.847	−1.551	4.132	**+1**
12	1.534	1.067	−1.991	1.037	0.334	3.499	**+1**
13	−0.479	0.371	0.941	0.398	0.893	3.391	**+1**
14	−0.633	−0.623	−0.492	1.442	2.314	4.372	**+1**
15	−1.046	1.920	−0.969	0.236	0.588	3.827	**+1**
16	1.620	1.127	1.491	−1.297	−0.861	3.459	**+1**
17	1.296	1.226	−0.415	−0.447	0.163	3.253	**+1**
18	−0.107	0.992	−0.598	0.024	1.206	3.363	**+1**
19	−1.002	1.383	0.837	−0.486	0.621	3.224	**+1**
20	0.959	1.631	0.885	−0.672	−1.017	3.434	**+1**
21	0.782	0.794	−0.044	−0.033	0.339	3.786	**+1**
22	0.579	0.685	−0.457	0.024	0.911	3.716	**+1**
23	−0.561	1.617	−0.006	−0.988	0.786	3.672	**+1**

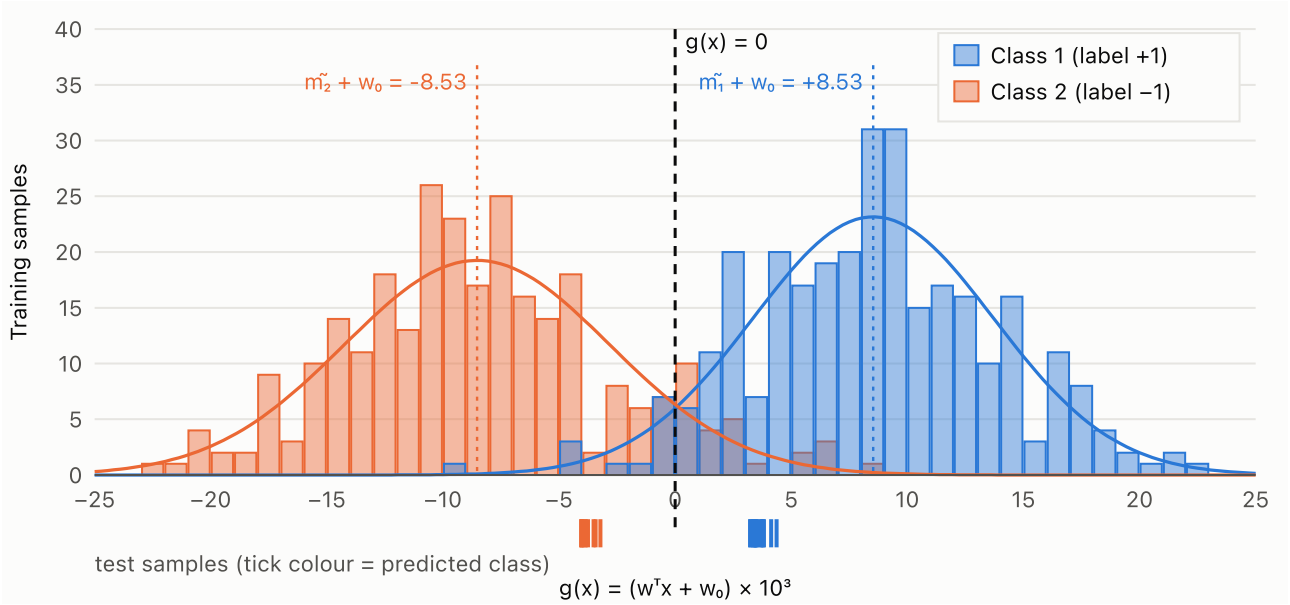


Figure 2: Training samples projected onto the Fisher direction, with a histogram of $g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$ per class (bin width 1×10^{-3}) and the fitted 1-D Gaussians drawn over them. The heavy dashed line is the boundary $g = 0$, the thin dotted lines mark the projected class means shifted by the bias, $\tilde{m}_1 + w_0$ and $\tilde{m}_2 + w_0$, at $\pm 8.53 \times 10^{-3}$, and the ticks below the axis are the 23 test samples.